



Investigating Fuel Efficiency of Heavy-Duty Vehicle Platooning Using a CFD Model

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Abstract

Platooning represents a crucial strategy for mitigating emissions from heavy-duty vehicles (HDVs). This study evaluates the effects of platoon composition on the surrounding airflow utilizing a computational fluid dynamics (CFD) model, and quantifies the resultant fuel efficiency and CO₂ emissions. This study examines fuel consumption data reconstructed from field experiments to validate the CFD model's ability to accurately simulate drag forces within a homogeneous three-truck platoon. The potential for fuel savings was assessed based on CFD-simulated fuel consumption, taking into account various inter-vehicle distances and driving speeds. The model successfully reproduced the fuel consumption observed in a platooning formation comprising lead, middle, and trailing trucks, with an error margin below 6.2%. Fuel consumption analysis shows that while lead and middle trucks consume more fuel with increased inter-vehicle distances, the trailing truck's consumption decreases at specific distance-to-length ratios (D/L), increasing again beyond a D/L of 1.1. Additionally, a significant decrease in total fuel efficiency was noted for D/L ratios exceeding 1.5. Considering the diverse platooning scenarios analyzed, the study anticipates an annual reduction of up to 7 tons of CO₂ equivalent per vehicle. By optimizing platooning configurations, this research contributes to enhancing fuel efficiency and reducing emissions from HDVs.

Keywords Heavy-duty vehicles · Truck platooning · CFD · Energy efficiency · Carbon dioxide emissions

1 Introduction

It is obvious that transportation sector should be considered as one of the most significant emission sources of both Green House Gases (GHGs) and air pollutants in urban and

road scales. Previous studies have demonstrated comprehensive efforts to combine the scientific findings with the engineering applications for enhancing the understanding for microscale turbulence and its aerodynamic impacts on flow characteristic by using computational fluid dynamics (CFD)-based approaches in urban and road-scale environments (Kim et al. 2001; Kim and Lee 2011; Sanchez et al. 2017; Santiago et al. 2017; Kadaverugu et al. 2019). In this regard, it would be desirable to establish a link between advanced transportation strategies for future and environmental and climatological issues in urban and road scales.

In recent years, advancements in transportation technology have significantly enhanced the convenience of goods and human mobility, contributing to an expansion of global interconnectedness (Transport Forum: ITF 2023). The transportation sector significantly contributes to global greenhouse gas (GHG) emissions, accounting for approximately 15% of the total (IEA 2023; IPCC 2022). Notably, heavy-duty vehicles (HDVs) play a critical role in the road transport sector's GHG emissions despite comprising a relatively small portion of the vehicle population (EEA 2023; EPA 2023). Diesel-powered HDVs are not only sources of GHGs

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but also emit air pollutants during operation (Grigoratos et al. 2019). Moreover, fuel efficiency in this sector remained stagnant before the implementation of HDV-related regulations (Muncrief and Sharpe 2015). In response to environmental challenges, numerous countries have enacted and strengthened regulations targeting CO₂ emissions standards for HDVs (e.g., European Commission 2019; ME 2020; Ragon et al. 2023). Consequently, enhancing the fuel efficiency of HDVs has emerged as a crucial strategy for reducing emissions in the road transport sector. This focus on maximizing fuel savings aligns with global efforts to address the climate crisis.

Intelligent transportation systems have facilitated the development of vehicle network technologies such as cooperative adaptive cruise control and vehicle-to-vehicle communication within the transportation system (e.g., Milanés et al. 2013; Wang et al. 2020). These technologies have sparked increasing interest in autonomous platooning systems. Platooning refers to a system where a following vehicle maintains a constant longitudinal and lateral distance behind the leading vehicle (Maiti et al. 2017; Bhoopalram et al. 2018). The platooning formation alters the wake shape of the downstream leading vehicle, impacting the frontal aerodynamics of the following vehicle and yielding benefits such as reduced fuel costs and improved energy efficiency (Kunze et al. 2010; Turri et al. 2016; Zhang et al. 2020; Pi et al. 2023; Cheng et al. 2023).

Fuel consumption in platooning varies based on factors including the number of vehicles in the platoon, vehicle speed, and the distance between vehicles. Several studies (e.g., Liang et al. 2015; Zhang et al. 2024) have examined fuel efficiency effects using two to four vehicles and various separation distances. In platooning systems, the separation between vehicles is a key factor influencing fuel consumption. Previous studies have used both wind tunnel and field experiments to investigate strategies for fuel savings in platooning vehicles (e.g., Browand et al. 2004; Tsugawa et al. 2011; Lammert et al. 2014; McAuliffe et al. 2017, 2018; Stegner et al. 2021). Field experiments provide a precise experimental environment that allows for repetitive testing. However, they encounter several challenges in measuring actual vehicle fuel efficiency. Limitations include limited control over real-world traffic and weather conditions, along with significant time, cost, and safety considerations (e.g., Salari and Ortega 2018; Zhang et al. 2024). To address these limitations, cost-effective numerical simulation methods have been proposed for the aerodynamic design of vehicles.

CFD has emerged as a critical tool for analyzing the aerodynamic efficiency of vehicle platooning, which can significantly reduce drag and fuel consumption. Recent studies (e.g., Tsugawa et al. 2011; Gnatowska and Sosnowski 2018; Scurtu and Gheres 2022) have demonstrated that vehicle platooning configurations, ranging from truck-only to mixed

setups, can reduce aerodynamic drag significantly. Truck platooning studies highlight energy savings through reduced drag at close spacings (Tsugawa et al. 2011), though very close distances can negate benefits (Gnatowska and Sosnowski 2018). Scurtu and Gheres (2022) emphasize trade-offs between aerodynamic efficiency and road safety, suggesting optimal spacing suitable for various situations. Despite these advancements, comprehensive studies on diverse platooning configurations and their extensive implications remain limited.

Smith et al. (2014) analyzed a two-truck platoon, reporting that the drag on the trailing truck decreased by up to 20% at an inter-vehicle distance of 30 feet, while the drag on the leading truck slightly increased. The overall drag reduction effect persisted up to a spacing of 160 feet, reaching a maximum of 23%. Vegendla et al. (2015) evaluated five platooning configurations involving 2–3 vehicles, revealing that single-lane in-line platooning offered the highest fuel savings, whereas side-by-side multi-lane configurations were disadvantageous in terms of drag. However, these studies focused primarily on limited configurations without exploring threshold values for optimal efficiency. Humphreys et al. (2016) analyzed fuel efficiency at varying inter-vehicle distances, noting that fuel savings increased at distances below 50 feet; however, the trailing vehicles experienced reduced efficiency due to buffering effects caused by vortices from the leading vehicle. While this study underscored the importance of spacing, it lacked a comprehensive analysis of multi-vehicle platooning. Luo et al. (2022) validated a numerical approach using the Reynolds-averaged Navier–Stokes (RANS)-based CFD model with a simplified passenger car geometry, the single 35° Ahmed body, and demonstrated that three-vehicle platooning achieves greater drag reduction compared to two-vehicle platooning. Although this study examined passenger cars rather than our focus on HDVs, it successfully demonstrated the reliability of CFD models and highlighted the enhanced drag reduction simulated in three-vehicle platooning compared to two-vehicle configurations. Siemon et al. (2018) analyzed four-vehicle platoons using CFD and TruckSim software simulations, revealing significant drag reduction at spacings of 30, 50, and 100 feet. However, the absence of experimental validation in their study limits the real-world applicability and reliability of the results.

Therefore, we address these limitations by integrating CFD simulations with experimental track tests to provide validated and actionable insights into three-vehicle platooning. Unlike previous studies, this research systematically examines seven distinct inter-vehicle spacings and identifies specific distance-to-length (D/L) ratios as thresholds for optimizing aerodynamic efficiency. By combining CFD simulations with experimental validation, we ensure robust and reliable results applicable to real-world platooning systems.

In this study, we aim to simulate the aerodynamic characteristics of a three-truck platoon using a CFD model that employs the RANS equations, which offer the advantage of shorter simulation times. We replicated the platooning system using the CFD model and validated it using track experimental measurements to ensure model reliability. Through this approach, we simulated and analyzed various scenarios involving different separation distances and vehicle speeds. Our goal is to quantify the reduction in CO₂ emissions resulting from fuel savings within a platoon, thereby demonstrating the significant impact of platooning on fuel consumption.

2 Methods

2.1 Field Experiment Data

To optimize the energy-saving potential of platooning, critical factors include inter-vehicle distance, vehicle weight, and vehicle speed. The separation distance between vehicles, in particular, has been identified as a crucial factor influencing fuel savings. McAuliffe et al. (2017) and McAuliffe et al. (2018) conducted research involving track-based fuel consumption tests, with the distance between trucks as a variable. McAuliffe et al. (2017) conducted their experiments in accordance with the SAE J1321 Type II fuel consumption testing procedure (SAE J1321). The SAE J1321 Type II Procedure is a standardized method used to measure fuel consumption in vehicles, particularly for evaluating the impact of platooning systems on fuel efficiency. This test involves gravimetric measurements of fuel consumption, which are highly accurate and reliable for assessing the effects of various conditions and configurations on platooning systems (Lammert et al. 2014, 2020; McAuliffe et al. 2020).

These experiments quantified the rate of fuel savings (ΔF_w) by measuring the average fuel consumption over a predetermined distance (Fig. 1(a)). To calculate fuel savings rates, the ratio of fuel weight used by the test trucks to that used by a control truck was utilized. In all trials,

control and test trucks were driven in tandem, with the control vehicle being an additional truck that replicated each configuration across all track test runs while maintaining a consistent distance under identical conditions. Two types of experiments were conducted: baseline and platoon tests. The baseline experiment was designed with vehicles spaced at one-quarter of the track length apart, ensuring that the three vehicles and one control vehicle did not interact with each other. The platoon experiments involved driving three vehicles in platoon formation, with the tests varying the separation distance and accordingly measuring fuel consumption. The ΔF_w was calculated using Eq. (1).

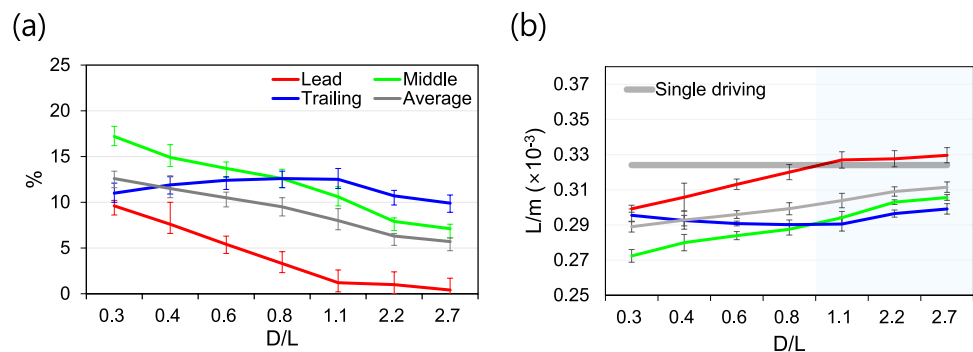
$$\Delta F_w(\%) = \frac{\overline{\Delta F_{w_b}} - \overline{\Delta F_{w_p}}}{\overline{\Delta F_{w_b}}} \quad (1)$$

Here, ΔF_{w_b} represents the fuel savings rate for the baseline test run with independent vehicles, and ΔF_{w_p} indicates the fuel savings rate for the platoon test runs with the three platoon vehicles. To convert fuel consumption rates into fuel efficiency values, diesel density was assumed to be 0.85 kg/L. The fuel weight measured from McAuliffe et al. (2017) was converted to L/km. Additionally, a consistency check between the converted and calculated fuel efficiency values from the test experiments was performed using Eq. (2).

$$\Delta F_{c_p}(\text{L/m}) = \overline{\Delta F_{c_b}} - \Delta F_w \cdot \overline{\Delta F_{c_b}} \quad (2)$$

Here, ΔF_{c_b} is the fuel consumption for the baseline test run, and ΔF_{c_p} is the fuel consumption for each truck (lead, middle, and trailing) in the platoon test runs. This study used ΔF_w values from McAuliffe et al. (2017) to evaluate experiments with varying inter-vehicle distances, assuming experiments were conducted using the same vehicles under similar environmental conditions. The $\overline{\Delta F_{c_b}}$ derived from McAuliffe et al. (2017) served as the reference, and ΔF_{c_p} for each test in McAuliffe et al. (2018) was calculated, as shown in Fig. 1(b).

Fig. 1 Measured **a** fuel saving rate (%) from SAE J1321 fuel consumption tests and **b** fuel consumption (L/m) calculated from fuel saving rate, depending on inter-vehicle distance in three-platoon test runs and single driving truck (Blue shading indicates the range where the fuel consumption of the lead truck and the single truck becomes similar)



2.2 Computational Setup

The CFD model employed in this study is a custom-developed research tool, specifically designed for simulating urban meteorological fields. This model utilizes a three-dimensional cubic grid system in the x , y , and z directions and is capable of conducting experiments and research under both idealized and real meteorological conditions. For idealized cases, specific meteorological settings such as wind or temperature fields can be applied, enabling the study of controlled scenarios. This model provides a flexible framework to investigate phenomena that could occur under actual meteorological conditions, thereby facilitating the study of complex atmospheric processes in various environments.

The CFD model used in this study assumes a three-dimensional, nonhydrostatic, nonrotating, incompressible airflow system. It operates within the framework of the RANS equations. The model employs the finite volume method and the semi-implicit method for pressure-linked equations for numerical approximations of the governing equations, which are solved within a staggered grid system (Patankar 1980). A renormalization group k – ε turbulence closure scheme is implemented for turbulence parameterization. To implement wall boundary conditions within the RANS equations, we utilize empirical formulations that account for the formation of the viscous sublayer, as described by Versteeg and Malalasekera (1995). These conditions include the employment of wall functions that capture the effects of the wall on turbulent flows. The empirical relationship used is expressed as follows:

$$y^+ = \frac{\Delta y_p}{\nu} C_\mu^{1/4} k^{1/2} \quad (3)$$

Here, Δy_p represents the distance from the near-wall node to the solid surface, ν is the kinematic viscosity. C_μ is an empirical constant (0.0845), k is turbulent kinetic energy. If the y^+ exceeds 11.63, the flow is considered turbulent, and the wall function approach is applied. In the near-wall turbulent flows, universal distribution of velocity (u^+) is calculated using the log-law formula:

$$u^+ = \frac{1}{\kappa} \ln(9.8y^+) \quad (4)$$

κ is the Von Karman constant (0.4). For comprehensive details on the CFD model applied, refer to Kim and Baik (2010) and Park et al. (2020).

The model's initial conditions for the wind field are based on a logarithmic function, assuming neutral atmospheric conditions. The initial wind components (U , V , W), turbulence kinetic energy (TKE, k), and TKE dissipation rate (ε) are set following the specifications by Castro and Apsley (1997) to establish the inflow conditions:

$$U(z) = \frac{u_*}{\kappa} \ln \frac{z}{z_0} \cos \theta, \quad (5)$$

$$V(z) = 0, \quad (6)$$

$$W(z) = 0, \quad (7)$$

$$k(z) = \frac{u_*^2}{C_\mu^{1/2}} \left(1 - \frac{z}{\delta}\right)^2, \quad (8)$$

$$\varepsilon(z) = \frac{C_\mu^{3/4} k^{3/2}}{\kappa z} \quad (9)$$

where u_* represents the friction velocity, z_0 is the roughness length (0.05 m), δ denotes the boundary-layer depth (set to 1000 m), and θ is the wind direction. In the simulation setup, inflow conditions are applied at the western boundary, while outflow conditions are designated at the eastern boundary. The lateral boundaries are treated as shear-free flows. At the outflow and upper boundaries, a zero-gradient condition is implemented. On solid surfaces, wall boundary conditions are implemented according to Versteeg and Malalasekera (1995). The CFD model integration extends to 3600 s with a timestep of 0.5 s to ensure the initial boundary condition incorporates a fully developed vertical profile.

2.3 Numerical Domain and Grid

The simulation of a three-truck platoon, drawn from a preceding study, was incorporated into the CFD model domain to replicate the experiment conducted by McAuliffe et al. (2018) (Fig. 2(a)). This platoon configuration consists of lead, middle, and trailing trucks. A Single driving test was also conducted, in which the single vehicle was driven alone. With a focus on horizontal resolution (x and $y=0.5$ m, $z=0.2$ m), the trucks were modeled with dimensions of 16 m in length (L), 2.5 m in width (W), and 4 m in height (H) for evaluation (Fig. 2(b)). The lead truck was positioned within the simulation domain at a distance of 40 m from the inflow boundary, with the domain itself extending 30 m in width and 24 m in height. The 3D grid structure between the trucks is depicted in Fig. 3(a), emphasizing the resolution in critical regions to accurately capture inflow and outflow characteristics. Figure 3(b) presents the YZ cross-section of the grid, highlighting the refined resolution near the trucks' edges to ensure precise representation of airflow interactions. The XY cross-section, as shown in Fig. 3(c), illustrates the arrangement of the three trucks and the grid spacing applied in the inter-truck regions. The separation distances for each case were aligned with those from the field experiment, though for cases EXP5 to EXP7, the separation distances in the

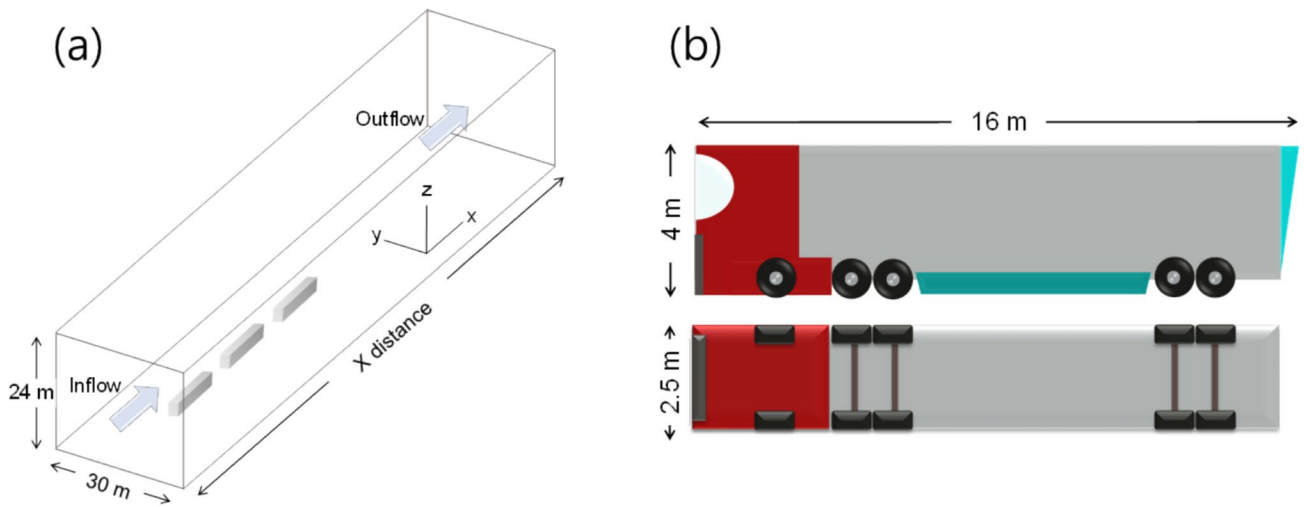


Fig. 2 Schematics of **a** computational domain and **b** dimensions of the trucks used for CFD modeling

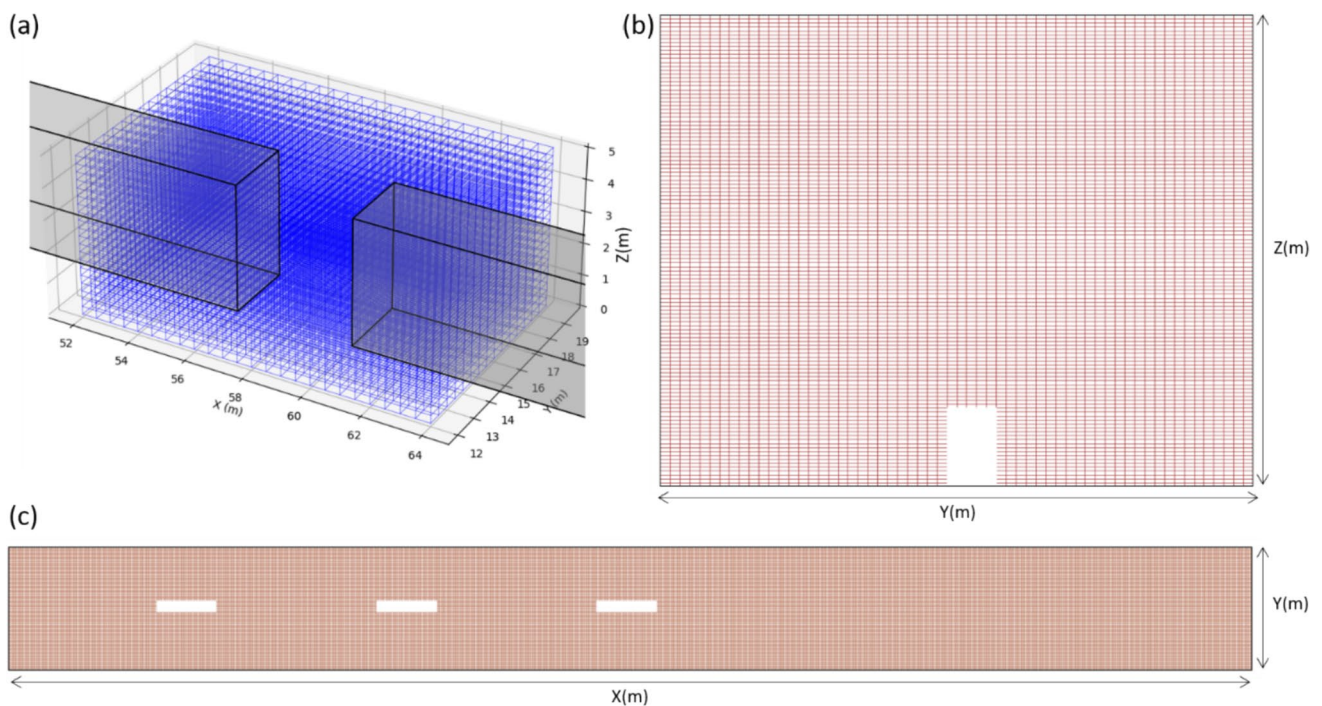


Fig. 3 Schematics of the Grid structure for CFD Modeling: **(a)** 3D Grid Between Trucks, **(b)** YZ Cross-section with Truck dimensions ($\Delta y = 0.5m$, $\Delta z = 0.2m$), and **(c)** XY Cross-section showing platoon positions at 4 m intervals (EXP1) starting from (40.0, 14.5)

numerical domain were adjusted by ± 0.1 m from those in the field due to the constraints of the minimum grid size that could be implemented (Table 1.)

Section 3.2.1 discusses simulations conducted to assess platooning effects based on truck length (L), replicating the conditions of McAuliffe et al. (2018) experiment. These simulations varied the length of the truck from two times (8 m) to five times (20 m) the height of the truck

(4 m) to evaluate the impact on drag force in different platooning formations. Section 3.2.2 discusses platooning experiments considering a range of driving speeds from 60 m/s to 120 m/s. The setup for driving-speed tests is the same as that for validation. In experiments examining truck configurations, the trailing truck was placed at a distance of 275 m (approximately 17H) from the outflow boundary.



Table 1 Dimensions of CFD model domain

EXP	Separation Distance (m)	D/L	kmax	jmax	imax	X-distance (m)
EXP1	4.0	0.3	64	64	672	336
EXP2	6.0	0.4	64	64	680	340
EXP3	9.0	0.6	64	64	692	346
EXP4	12.0	0.8	64	64	704	352
EXP5	17.5	1.1	64	64	726	353
EXP6	35.0	2.2	64	64	796	398
EXP7	43.5	2.7	64	64	834	417

2.4 Drag Calculations

The drag force exerted on vehicles is primarily composed of two components: skin friction drag and pressure drag. Skin friction drag is influenced by the vehicle's surface roughness and area, whereas pressure drag is determined by the shape of the wake formed as airflow separates from the leading vehicle and reattaches downstream. At sufficiently high Reynolds numbers, the aerodynamic drag on each vehicle is predominantly influenced by the pressure differential across its front and rear surfaces (Tsuei and Savas 2001; Bitog et al. 2011; Jurco 2021). The formation of a platoon can reduce the frontal dynamic pressure on subsequent vehicles, making the drag more dependent on pressure differences than on skin friction (Hummel 1995; Belzile et al. 2012).

In this study, the relationship between fuel consumption and drag force is examined using CFD-simulated pressure differentials and wind speeds. The total drag force (D_t) is defined as the sum of pressure drag (D_p) and friction drag (D_f) (Mohamed et al. 2015). Initially, the pressure drag force is computed as:

$$D_p = C_{D_p} \times 0.5 \times \rho \times U^2 \times A_F \quad (10)$$

The coefficient of pressure drag (C_{D_p}) is calculated as:

$$C_{D_p} = \frac{\Delta P}{(0.5 \times \rho \times U^2)} \quad (11)$$

Here, ρ represents the flow density, U is the driving speed, A_F is the frontal area of the truck, and ΔP is the pressure difference. Then, the friction drag force (D_f) is then calculated for the truck's roof and side areas as follows:

$$D_f = C_{D_f} \times 0.5 \times \rho \times U^2 \times A_{RS} \quad (12)$$

The coefficient of friction drag (C_{D_f}) is calculated as:

$$C_{D_f} = \frac{0.031}{(Re_L^{1/7})} \quad (13)$$

Here, A_{RS} is the combined area of the truck's roof and sides, Re is the Reynolds number, derived using the truck's length and the flow's kinematic viscosity (White 2016).

The total drag force (D_t) is subsequently utilized to estimate fuel consumption (F_c) as follows:

$$F_c = D_t \times \frac{BSFC}{\eta \rho_{fuel}} \quad (14)$$

Here, η denotes transmission efficiency, ρ_{fuel} is the diesel fuel density, and BSFC is brake-specific fuel consumption (Table 2).

3 Results

3.1 Evaluation

3.1.1 Evaluation with Measured Fuel Consumption (F_c)

This study utilized the fuel consumption data from McAuliffe et al. (2018) to assess the accuracy of CFD-simulated fuel consumption. The measured fuel consumption for both lead and middle trucks showed an increase as the

Table 2 Parameters used in estimating drag force and fuel consumption

Parameter	Value (unit)
Frontal area of truck	$A_f = 10 \text{ m}^2$
Side area of truck	$A_{rs} = \text{varies according to length of truck (m}^2\text{)}$
Vehicle speed	$V = 105 \text{ km h}^{-1} (29.17 \text{ m s}^{-1})$
Acceleration due to gravity	$g = 9.81 \text{ m s}^{-2}$
Transmission efficiency	$\eta = 0.82$
Air density	$\rho = 1.225 \text{ kg m}^{-3}$
Diesel Fuel density	$\rho_{fuel} = 850 \text{ kg m}^{-3}$
Kinematic viscosity	$\nu = 0.2 \times 10^{-3}$
BSFC (Brake-specific fuel consumption)	$6.76 \times 10^{-8} \text{ kg W}^{-1} \text{ m}^{-3}$

inter-vehicle distance increased (Table 3). Specifically, the lead truck's fuel consumption increased from 0.299 L/km to 0.330 L/km as the inter-vehicle distance widened (Fig. 1(b)). In EXP5, the lead truck's fuel consumption is 0.327 L/km, surpassing the fuel consumption of the single driving truck (0.324 L/km). From EXP5 onwards, there was not a significant change in fuel consumption despite an increase in D/L. Additionally, as D/L widened, the middle truck's fuel consumption also increased from 0.272 L/km to 0.306 L/km but remained lower than when single driving. The trailing truck exhibited the most significant fuel savings at a 12 m (EXP4) inter-vehicle distance, with fuel consumption decreasing from 0.295 L/km in EXP1 to 0.290 L/km in EXP4 and EXP5 before rising to 0.299 L/km in EXP7.

The lead truck faced the highest drag force according to CFD-simulated variables, with over 95% of the total drag force arising from the pressure drag, which increased alongside the inter-vehicle distance, leading to a rise in the lead truck's total drag force (Fig. 4(a)). As the inter-vehicle distance increased, the pressure drag force on the middle truck similarly rose, while the friction drag force declined until EXP4, then increased again until EXP7. The middle truck's total drag force mirrored the friction drag force trend. Meanwhile, the trailing truck exhibited a decrease in pressure drag force with an increasing inter-vehicle distance, and the friction drag force followed a decrease until EXP4, then an increase. The trend in total drag force resembled that of the friction drag force. Notably, although the friction drag force constituted only 4.2% to 6.0% of the total drag force, its pattern was influenced by the total drag force trend.

The CFD-simulated fuel consumption exhibited a trend analogous to the drag force for all three trucks (Fig. 4(b)). Fuel consumption for the lead and middle trucks increased with the inter-vehicle distance. Specifically, the lead truck's fuel consumption rose from 0.298 L/km to 0.328 L/km as the D/L ratio widened. In EXP6, the fuel consumption is 0.324 L/km, surpassing the measured fuel consumption of the single driving truck (0.324 L/km) and closely resembling the modeled fuel consumption of the single driving truck (0.336 L/km) (Table 4). From EXP6, the lead truck's fuel consumption follows

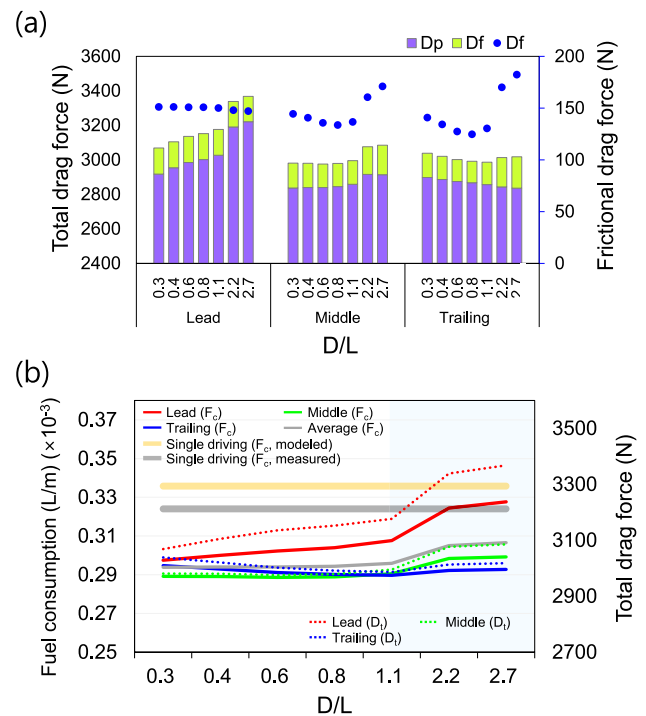


Fig. 4 CFD-simulated **a** total drag force, which is the sum of pressure and frictional drag force (represented by bar) and frictional drag force (indicated by black dot), and **b** fuel consumption (lead, middle and trailing, as well as measured and modeled single driving trucks), along with the total drag force depending on D/L (Blue shading same as Fig. 1(b))

the trend of measured fuel consumption. Additionally, as D/L increased, the fuel consumption of the middle truck also increased from 0.289 L/km to 0.299 L/km. The trailing truck's simulated fuel consumption decreased from EXP1 to EXP5 and then increased in EXP6 and EXP7. The CFD-simulated fuel consumption revealed an error margin within 6.1% for all inter-vehicle distance scenarios among the three trucks, indicating the CFD model's proficiency in replicating the fuel consumption observed in McAuliffe et al. (2018). Consequently, the CFD model is deemed reliable for predicting fuel consumption relative to platoon formation.

Table 3 Reconstructed fuel consumption data from SAE J1321 field experiment (McAuliffe et al. (2018))

EXP	Separation Distance (m)	Fuel consumption (L/km)			
		Lead Truck	Middle Truck	Trailing Truck	Average
EXP1	4.0	0.299	0.272	0.295	0.289
EXP2	6.0	0.306	0.280	0.292	0.293
EXP3	9.0	0.313	0.284	0.291	0.296
EXP4	12.0	0.320	0.288	0.290	0.299
EXP5	17.4	0.327	0.294	0.290	0.304
EXP6	34.9	0.328	0.303	0.296	0.309
EXP7	43.6	0.330	0.306	0.299	0.311



Table 4 CFD-simulated fuel consumption data

EXP	Separation Distance (m)	Fuel consumption (L/km)			
		Lead Truck	Middle Truck	Trailing Truck	Average
EXP1	4.0	0.298	0.289	0.295	0.294
EXP2	6.0	0.300	0.289	0.293	0.294
EXP3	9.0	0.302	0.289	0.291	0.294
EXP4	12.0	0.304	0.289	0.290	0.294
EXP5	17.5	0.308	0.291	0.290	0.296
EXP6	35.0	0.324	0.298	0.292	0.305
EXP7	43.5	0.328	0.299	0.293	0.307

3.1.2 Analysis of Pressure/Wind Speed Distribution

The findings from Section 3.1.1 highlighted that drag force arises from both pressure and friction, with pressure drag—significantly influenced by air pressure changes—predominantly affecting the overall drag force. To further explore vehicle aerodynamics, we analyzed the CFD-simulated pressure field. The pressure field was analyzed using the x – y cross-section, vertically averaged to the height of the trucks (Fig. 5), and the x – z cross-section, averaged across the truck width (Fig. 6). The observed pressure changes around the vehicles predominantly showed a negative distribution, with the changes being more pronounced at the rear than at the front of each vehicle. Notably, in the x – z cross-section, a significant pressure reduction was observed on the upper surface of the trucks.

Notably, the air pressure change at the rear of the lead truck decreased substantially as the distance between vehicles increased, marking the most significant reduction among the three trucks. In the x – y plane, while the pressure changes at the front of the lead truck remained relatively consistent, the pronounced decrease at the rear led to an increased pressure differential, thereby enhancing aerodynamic drag (Fig. 5). In the x – z plane, as the inter-vehicle distance increased, the pressure gradient changes were observed not only at the rear of the lead truck but also on its upper side (Fig. 6). This increase in drag led to greater fuel consumption as the inter-vehicle distance widened. In EXP7, the pressure changes at the rear of the lead truck was similar to that of the single truck (Fig. 5(d) and 6(d)). Based on the previous findings, fuel consumption increased when D/L exceeded 1.1, approaching that of the single truck. The

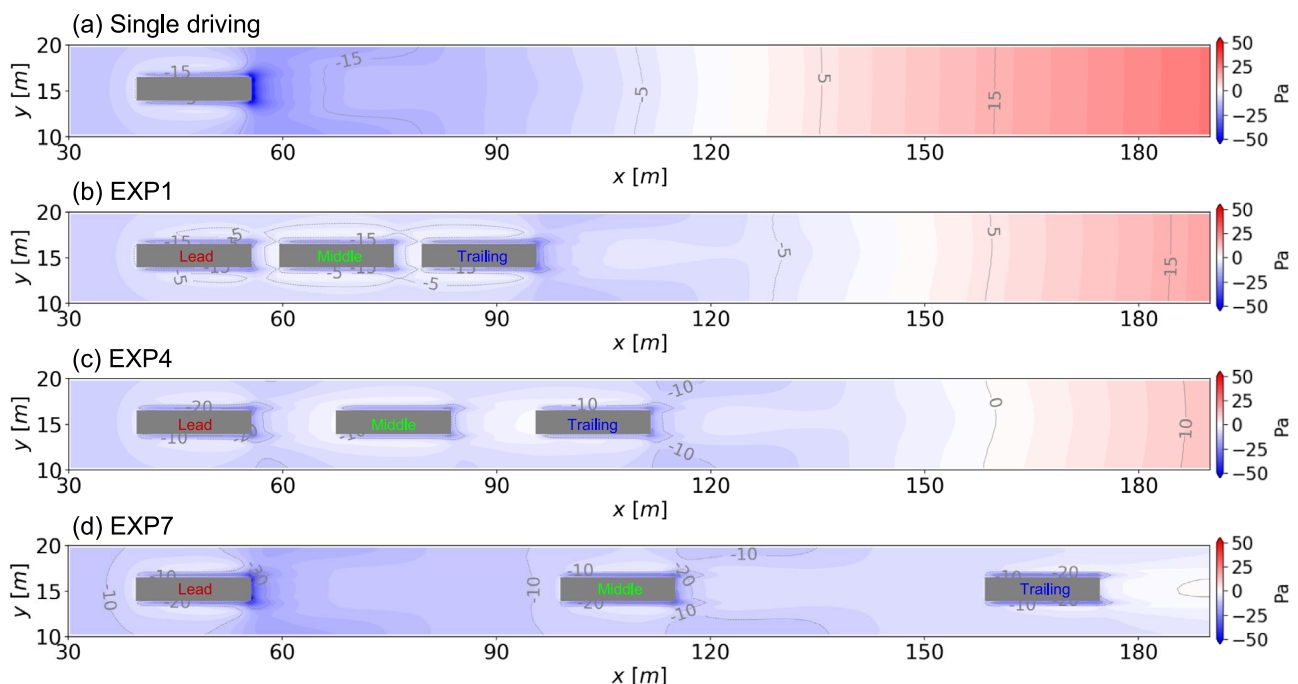


Fig. 5 The xy cross-section of CFD-simulated pressure gradient averaged up to truck height for **a** single driving truck, **b** EXP1, **c** EXP4, and **d** EXP7

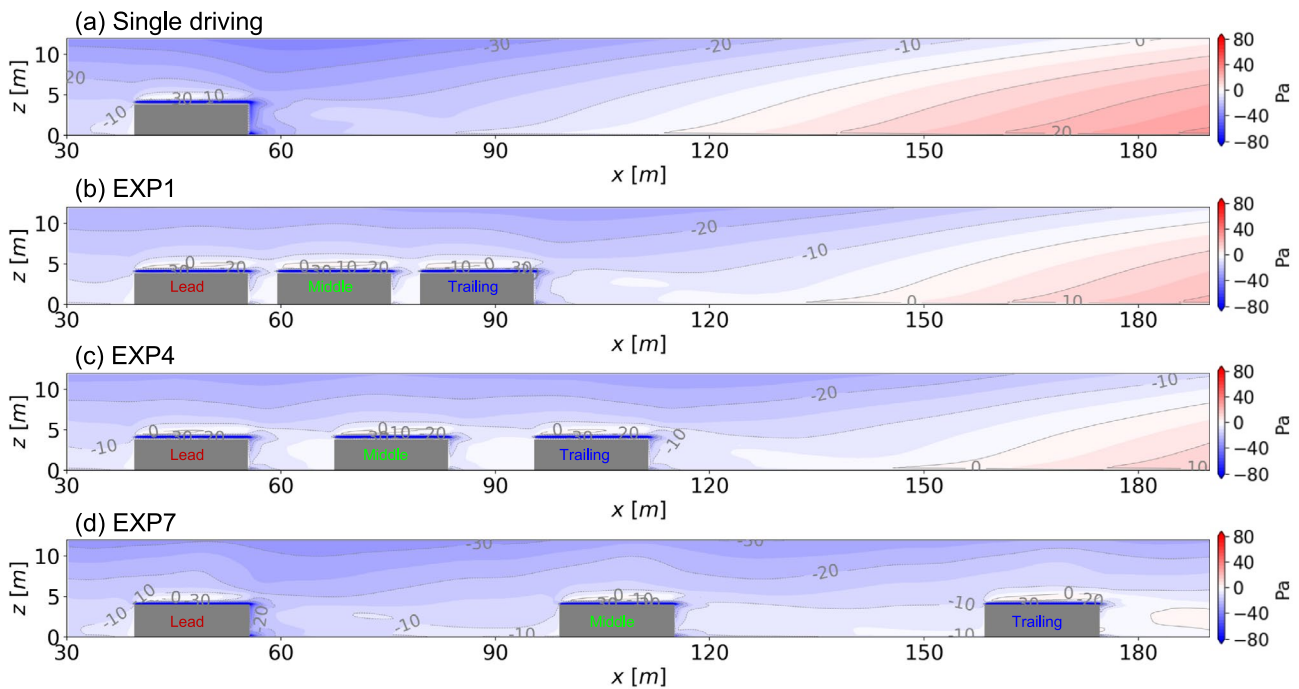


Fig. 6 The xz cross-section of CFD-simulated pressure gradient averaged across the truck width for **a** single driving truck, **b** EXP1, **c** EXP4, and **d** EXP7

distribution of pressure variation at the rear of the lead truck also becomes similar to that of a single truck when D/L exceeds 1.1.

Furthermore, as the inter-vehicle distance increased, a pressure drop was observed between the trucks. In EXP4, the pressure variation at the front of the middle and trailing trucks was the smallest, while the region affected by pressure changes was the widest. At the rear of the middle and trailing trucks, the influence of the upper low-pressure region became more pronounced with increasing inter-vehicle distance, resulting in a larger pressure drop compared to the front. These findings indicate that when the distance-to-length ratio (D/L) exceeded 1.1, the amplified pressure drop significantly increased aerodynamic drag. Accordingly, EXP4 highlights the optimal distance-to-length ratio. Ultimately, these trends in pressure variation had a direct impact on fuel consumption.

The analysis also extended to wind speed variations alongside pressure as key variables in drag force calculation. In the x–y cross-section, wind speeds were generally higher at the rear of the vehicles than at the front (Figure 7). The wind speed remained stable at the front of the lead truck but increased steadily at the rear as the distance between trucks increased. Similar to the distribution of pressure variation, in EXP7, the wind speeds at the rear of the lead truck were similar to those of the single truck.

However, wind speeds around the middle and trailing trucks did not show a direct proportionality to the

inter-vehicle distance. For inter-vehicle distances between 12 and 17 m (EXP4, EXP5), the wind speeds were lowest across all points, particularly at the fronts of the middle and trailing trucks within this distance range. The wind speed was notably lower at the fronts than the rears of these trucks, reflecting the pattern observed in pressure changes—larger at the vehicle's rear than at the front. Such variations in wind speed significantly influenced the frictional drag force, thereby affecting the middle and trailing truck's fuel consumption (The dot plot in Fig. 4(a)). These observations affirm that the CFD model employed in this study effectively reproduces the measured fuel consumption trends according to vehicle spacing.

3.2 Fuel efficiency (F_e)

3.2.1 Ratio of Length to Inter-Vehicle Distance (D/L) Scenarios

In the analyses presented in Section 3.1, we examined variations in drag force and fuel efficiency with different inter-vehicle distances using three trucks, maintaining a truck length of 16 m. However, vehicles come in diverse shapes and sizes on the road, and thus, not only the distance between vehicles but also the shape of the vehicle utilized for platooning can significantly impact aerodynamics and fuel efficiency. Given the constraints of road width due to traffic regulations, we modified the length of the vehicle to

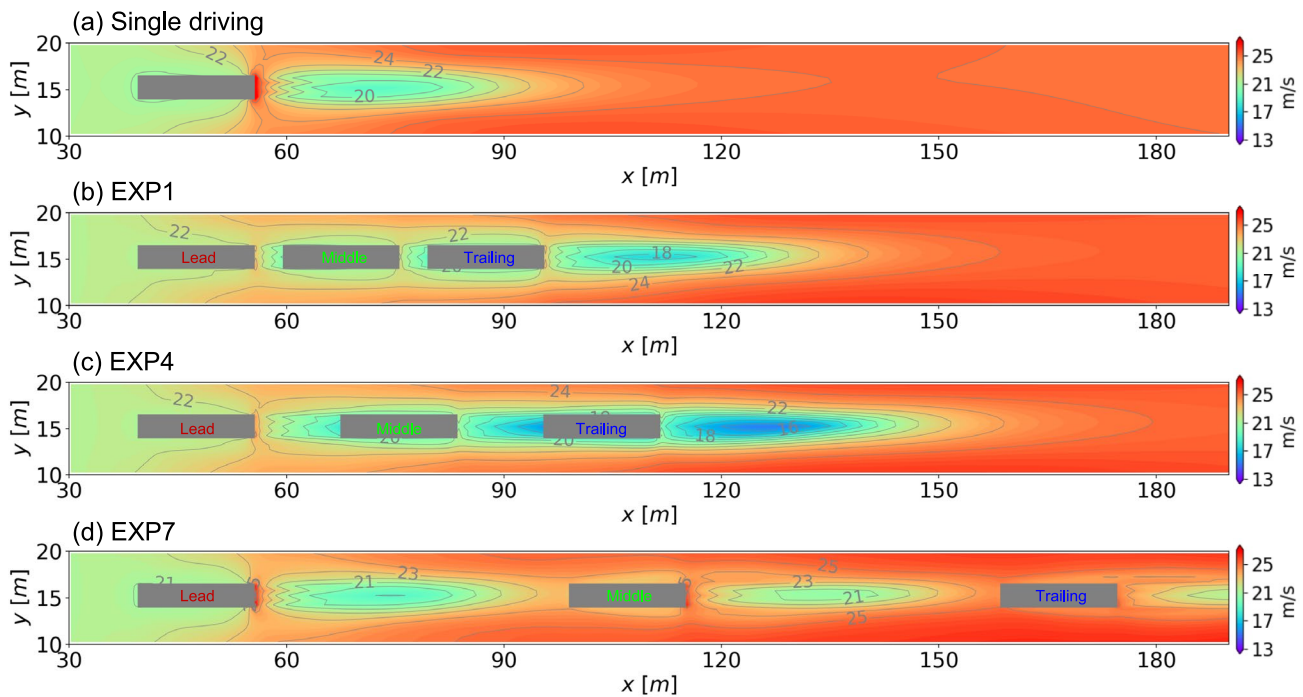


Fig. 7 The xy cross-section of CFD-simulated wind speed averaged up to truck height for **a** single driving truck, **b** EPX1, **c** EXP4, and **d** EXP7

assess the pattern of fuel consumption across various vehicle shapes. This examination involved analyzing the change in fuel efficiency based on the ratio (D/L) of the distance between vehicles (D) to the length (L) of the truck. D was set for seven experiments (ranging from 4.0 m to 43.6 m), as assumed in the evaluation, with the truck's length altered based on the vehicle's height (H), spanning from two ($R2$, $L = 8$ m) to five times ($R5$, $L = 32$ m) the height. A comprehensive set of 28 experimental cases was conducted from $R2$ to $R5$ (Appendix Table 7).

As confirmed by the evaluation results, the fuel efficiency of the lead truck was particularly sensitive to changes in the D/L ratio (Fig. 8(a)). The average fuel efficiency recorded was 3.24 km/L, marking the least efficiency when compared to the middle and trailing trucks. A distinct pattern emerged, showing a decrease in fuel efficiency with an increase in D/L , particularly noting a marked decline at $D/L < 1.5$. At a closer inter-vehicle distance ($D/L < 1.5$), the average fuel efficiency was 3.32 km/L, which dropped to 3.08 km/L (a decrease of 0.24 km/L) for $D/L > 1.5$. The shortest vehicle length, denoted as $R2$, demonstrated superior fuel efficiency for $D/L < 1.5$, indicating a more pronounced decrease in fuel efficiency as D/L increased.

For middle trucks, the overall average fuel efficiency was recorded at 3.44 km/L, exhibiting the highest fuel efficiency among the three vehicles. The shorter the length of the truck, the higher the fuel efficiency observed (Fig. 8(b)). Although fuel efficiency gradually decreased with an increase in D/L ,

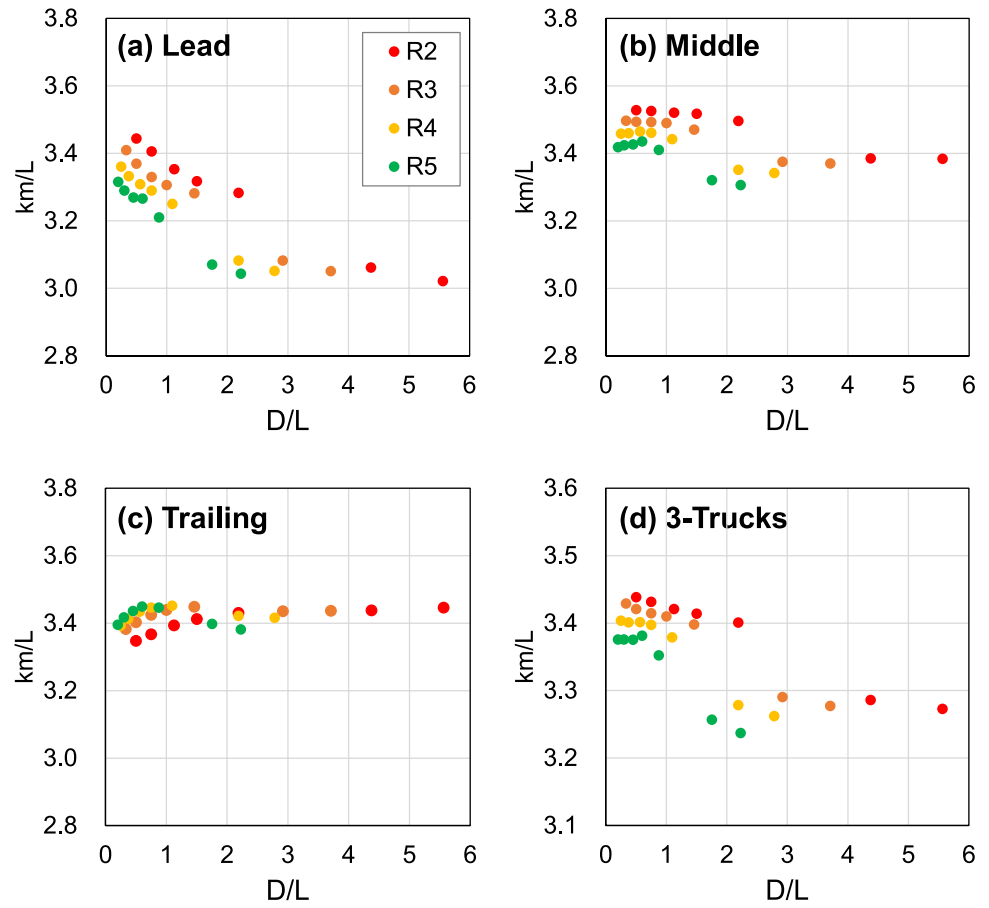
a unique trend was noted where $R4$ and $R5$ exhibited an initial increase, followed by a decrease when D/L was less than 1.5. Similar to the lead trucks, fuel efficiency for the middle trucks also tended to decline past the threshold of $D/L = 1.5$. In the case of the trailing truck, aside from $R2$, a pattern emerged where fuel efficiency first increased and then decreased as D/L rose, with $R2$'s fuel efficiency increasing in relation to D/L (Fig. 8(c)). Contrary to the lead and middle trucks, the fuel efficiency changes across the three trucks were more continuous, exhibiting relatively minor differences as D/L varied.

The fuel efficiency across the three trucks ranged from a maximum of 3.44 km/L ($D/L = 0.5$) to a minimum of 3.24 km/L ($D/L = 2.23$) (Fig. 8(d)). $R2$ showed superior fuel efficiency, indicating relatively higher efficiency for shorter trucks compared to longer ones. A noteworthy pattern emerged, highlighting a sharp decrease in fuel efficiency when D/L exceeded 1.5. This pattern suggests that, on average, higher fuel efficiency is observed when the ratio of vehicle length to inter-vehicle distance is below 1.5.

3.2.2 Vehicle Speed Scenarios

Further investigation is required to comprehensively understand the complex aerodynamic interactions that occur among platoon tractor-trailer combinations at highway speeds. In scenarios of conventional single-truck driving, an increase in driving speed results in heightened drag,

Fig. 8 Fuel efficiency depending on ratio of length to distance (D/L) for **a** lead truck, **b** middle truck, **c** trailing truck, and **d** average across all three trucks, and depending on ratio of height to length of truck (L/H) for R2 (red), R3 (orange), R4 (yellow), and R5 (green)



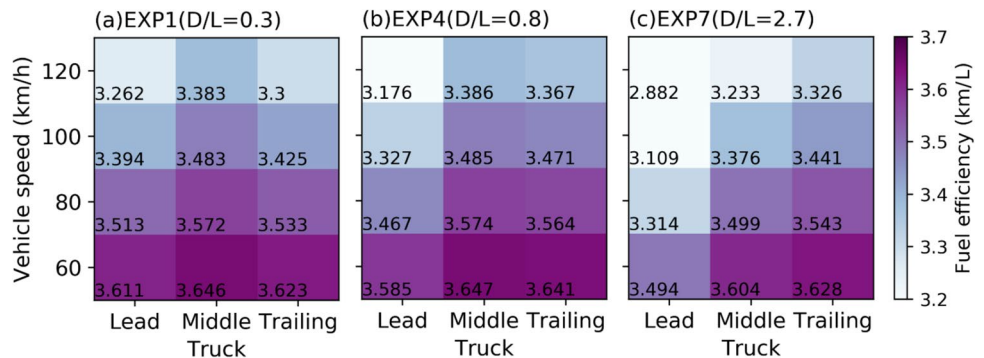
which, in turn, leads to increased fuel consumption. This study explored the variation in fuel efficiency of each vehicle within a platoon with the increase in platoon speed from 60 km/h to 120 km/h, considering the inter-vehicle distance.

The experimental results reveal that fuel efficiency declines for all vehicles as the driving speed increases, irrespective of the inter-vehicle distance (Fig. 9). Additionally, a relatively consistent reduction in fuel efficiency was observed across all vehicles with an increase in driving speed. Nonetheless, the specific impact on each vehicle varied depending on the scenario. For instance, in EXP1

(D/L=0.3), featuring the closest inter-vehicle distance, the disparity in fuel efficiency among the three trucks was minimal as the driving speed increased (Fig. 9(a)). At the highest evaluated driving speed of 120 km/h, the fuel efficiency difference between the middle truck (the highest) and the lead truck (the lowest) was 0.121 km/L.

In EXP4 (D/L=0.8), the decline in fuel efficiency for the lead truck was more marked as the driving speed increased (Fig. 9(b)). The increase in the D/L ratio from 0.3 to 0.8 resulted in a decrement of 0.026 km/L in fuel efficiency for the lead truck at 60 km/h, which further decreased by 0.086 km/L

Fig. 9 Fuel efficiency based on inter-vehicle distances and vehicle speed for **a** lead truck, **b** middle truck, and **c** trailing truck



at 120 km/h. Moreover, at a D/L ratio of approximately 0.8, the reduction in fuel efficiency for both the middle and trailing trucks with increasing driving speed was relatively similar.

As the D/L ratio increased, the lead truck experienced a more significant decline in fuel efficiency with an increase in driving speed. Particularly, in EXP7 (D/L = 2.7), where the inter-vehicle distance was at its maximum, the decrease in fuel efficiency was consistent across speeds. However, the variance in fuel efficiency among the three trucks was most pronounced (Fig. 9(c)). At 120 km/h, the fuel efficiency difference between the highest (trailing truck) and the lowest (lead truck) reached 0.444 km/L, indicating a decrease approximately 3.67 times greater than that observed at D/L = 0.3. Notably, at D/L = 2.7, the fuel efficiency of the lead truck decreased by 0.611 km/L as the driving speed doubled, approximately 1.8 times the decrease observed at D/L = 0.3.

The study elucidated that operating in a platoon formation, lower driving speeds and reduced inter-vehicle distances contribute to enhanced fuel efficiency. However, aligning with the evaluation results, the fuel efficiency of the trailing truck improved until reaching a D/L of 0.8, thereafter showing a downward trend. The explored scenarios spanned inter-vehicle distances from 4 m to 43.5 m, representing an approximately 10.9-fold difference, and driving speeds from 60 km/h to 120 km/h across four scenarios, marking a twofold variation. While fuel efficiency exhibited a maximum variance of 0.179 km/L based on inter-vehicle distance, a significant alteration of 0.439 km/L was noted with changes in vehicle speed. According to this study's scenarios, the average fuel efficiency across the three trucks showed a higher sensitivity to variations in driving speed than to changes in inter-vehicle distance, notably within the lead truck. This analysis enables the quantification of changes in fuel efficiency in response to variations in driving speed (Fig. 10). The equation for calculating fuel efficiency based on wind speed according to inter-vehicle distance is as follows.

$$y = -0.0052x + 3.9478 \quad (15)$$

$$y = -0.0073x + 4.024 \quad (16)$$

where x represents the vehicle speed, y is the fuel efficiency. Equation (15) represents EXP1 (D/L = 0.3), whereas Eq. (16) corresponds to EXP7 (D/L = 2.7).

3.3 CO₂Emissions

Numerous nations, including the US, the EU, and Canada, among others, are implementing stricter regulations on greenhouse gas emissions from HDVs to transition towards a carbon-neutral society. This shift involves moving from traditional fuel-based internal combustion engines to eco-friendly vehicles that emit no greenhouse gases. Current

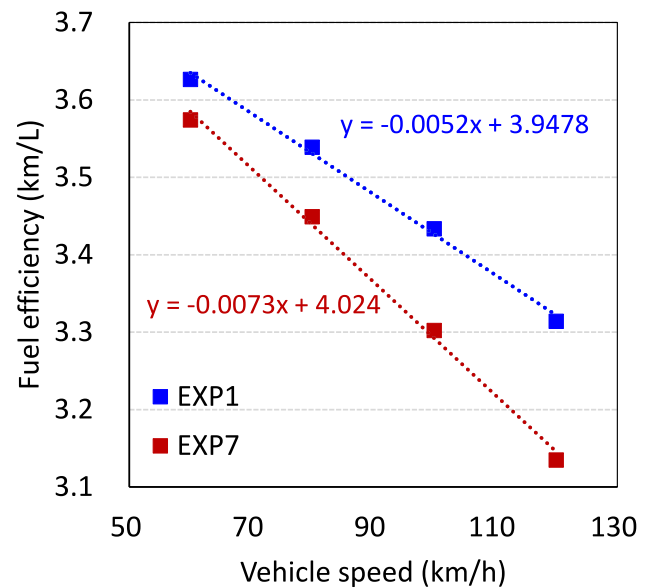


Fig. 10 Fuel efficiency depending on vehicle speed for EXP1 (D/L = 0.3) and EXP7 (D/L = 2.7)

advancements in platooning technology are projected to enhance energy efficiency significantly, contributing to the reduction of greenhouse gas emissions during this transition period (Massar et al. 2021; Paddeu and Denby 2021; Miklautsch and Woschank 2022; Zhang et al. 2024). This study's CFD-simulated results indicate that fuel efficiency can vary with different platooning configurations. Using these simulated results, our aim is to estimate CO₂ emissions employing the Tier 1 methodology as delineated in the Revised 2006 IPCC Guidelines (IPCC 2006). The formula for computing the transportation sector's GHG emissions is as follows:

$$Emission = \sum_a [F_c \cdot EF_a] \quad (17)$$

where CO₂ emissions are determined by the product of fuel consumption (F_c) (in liters) and the emission factor (EF_a) (in kg/TJ) for each specific fuel type (a). Given the net calorific value of diesel at 35.3 MJ/L and an emission factor of 74100 kg/TJ (IPCC 2006), emissions are calculated as:

$$Emission = x \text{ L} \times 35.3 \left(\frac{\text{MJ}}{\text{L}} \right) \times 10^{-6} \left(\frac{\text{TJ}}{\text{MJ}} \right) \times 74100 \left(\frac{\text{kg}}{\text{TJ}} \right) \\ = 2.61573 \times x \text{ kgCO}_2 - eq \quad (18)$$

To accurately compute emissions, the distance HDVs travel, which varies by country, must be considered. For this study, we base our estimates on data from South Korea, where HDVs averaged a daily travel distance of 135.1 km from 2015 to 2020, according to national statistics (Seo and Park 2023).

Table 5 CO₂ emissions derived from CFD-simulated fuel consumption data, considering various D/L scenarios

Scenario	D/L	Annual Used fuel (L/year)	CO ₂ emissions per year (tons CO ₂ -eq per year)
D/L	0.5 (Min)	43022.6	112.54
	2.23 (Max)	45699.8	119.54
	D/L < 1.5	14498.7	113.8
	D/L > 1.5	15012.9	117.8

Table 6 Difference between maximum and minimum CO₂ emissions derived from CFD-simulated fuel consumption data for various vehicle speed scenarios

Scenario	D/L	Annual Used fuel (L/year)	CO ₂ emissions per year (tons CO ₂ -eq per year)
Vehicle speed	0.3	3847.37	10.06
	0.8	3921.13	10.26
	2.7	5796.70	15.16

Analysis from Section 3.2.1 revealed the minimum and maximum fuel efficiency for the three vehicles, based on the ratio of inter-vehicle distance to vehicle length, to be 3.24 km/L (D/L = 2.23) and 3.44 km/L (D/L = 0.5), respectively. Utilizing this statistical data, the difference in fuel consumption based on D/L ratios amounts to 2677.2 L/year, equivalent to approximately 19.2 kg CO₂-eq, resulting in an approximate annual reduction of 7 tons CO₂-eq. Dividing the average values for D/L by the threshold of 1.5, the CO₂ emissions are calculated at 14,499 L/year for D/L < 1.5 and 15,013 L/year for D/L > 1.5, presenting a difference of approximately 11 kg CO₂-eq, or 4 tons CO₂-eq per year (Table 5).

From the data in Section 3.2.2, the variance in fuel consumption due to speed changes within the same inter-vehicle distance ranges from 10.5 L (D/L = 0.3) to 15.9 L (D/L = 2.7). This variance could lead to an annual CO₂-eq reduction ranging from a minimum of approximately 10.0 tons to a maximum of 15.2 tons (Table 6).

Optimizing platooning operation conditions by utilizing the fuel efficiency insights gained from this study, including considerations of vehicle speed and travel distance, can play a pivotal role in advancing toward a carbon-neutral society through effective vehicle operation strategies.

4 Concluding Remarks

This study employed a CFD model to simulate the aerodynamic characteristics of vehicle platooning. Through the comparison and validation of CFD-simulated results with track experiment

data from existing literature, we proposed an optimal inter-vehicle distance for enhancing drag force and fuel efficiency. An innovative aspect of this research was its application of a CFD model, typically used in assessing the meteorological environment around buildings, to model vehicle platooning configurations.

The evaluation revealed that the maximum error in CFD-simulated fuel consumption for the lead, middle, and trailing trucks compared to measured values in all tested inter-vehicle distances was 6.15%. This accuracy indicates the CFD model's capability to reproduce the observed reduction in fuel consumption as inter-vehicle distance increases. When D/L ratio was greater than 1.1, the fuel consumption of the lead truck gradually increased, becoming similar to the fuel consumption of the single driving truck. Notably, the trailing truck demonstrated the lowest fuel consumption at a D/L ratio of 1.1, aligning with measured data and underscoring the model's efficacy in simulating fuel consumption scenarios with a simplified platooning truck configuration.

Further analysis of the distribution of pressure and wind speed indicated that over 94% of the drag force was attributable to pressure differential. However, for the trailing truck, surface friction changes with varying inter-vehicle distances had a noticeable impact on fuel consumption. The use of a CFD model for platooning simulation provided insights into the underlying causes of fuel consumption changes observed in road track experiments.

Fuel efficiency varied significantly across the three trucks depending on the D/L ratio. Based on a comprehensive range of D/L scenarios, the simulation results exhibited distinct patterns of fuel efficiency changes for the lead and middle trucks, with a notable decrease in fuel efficiency for D/L ratios greater than 1.5. Additionally, the fuel efficiency across platooning configurations ranged from 3.24 km/L (D/L = 2.23) to 3.44 km/L (D/L = 0.5) on average per liter of fuel. Using Korean statistical data, we infer that different platooning formations could result in an annual CO₂ equivalent reduction of approximately 7 tons per HDV, with potential savings of up to 15.2 tons of CO₂ equivalent annually, depending on vehicle speed. Based on the present results of fuel efficiency variations under the platooning form scenarios, optimal efficiency can be predicted, even with other forms of vehicles.

This research contributes to the ongoing discourse on platooning's benefits by offering insights into optimizing fuel efficiency and reducing CO₂ emissions through varied platooning formations. It emphasizes the potential impact of HDV platooning configurations on CO₂ emissions, as modeled through CFD simulations. Future studies should consider diverse factors such as traffic conditions, infrastructure, and vehicle specifics to expand our understanding of platooning technology's role in reducing CO₂ emissions. Overall, platooning technology is poised to play a crucial part in devising strategies for achieving a carbon-neutral transportation sector.

Appendix

Table 7 Fuel efficiency (km/L) based on ratio of length to distance (D/L) for lead truck, middle truck, trailing truck, and average of three trucks

R	EXP	D/L	Fuel efficiency (km/L)			
			Lead Truck	Middle Truck	Trailing Truck	Average
R2	EXP1	0.50	3.444	3.528	3.348	3.439
	EXP2	0.75	3.406	3.526	3.368	3.432
	EXP3	1.13	3.353	3.520	3.393	3.421
	EXP4	1.50	3.317	3.518	3.413	3.414
	EXP5	2.19	3.283	3.496	3.431	3.401
	EXP6	4.38	3.062	3.385	3.438	3.286
	EXP7	5.56	3.021	3.384	3.446	3.273
R3	EXP1	0.33	3.410	3.497	3.383	3.429
	EXP2	0.50	3.369	3.493	3.403	3.421
	EXP3	0.75	3.330	3.493	3.425	3.415
	EXP4	1.00	3.306	3.490	3.439	3.410
	EXP5	1.46	3.281	3.470	3.449	3.398
	EXP6	2.92	3.082	3.375	3.436	3.290
	EXP7	3.71	3.051	3.370	3.437	3.277
R3	EXP1	0.25	3.361	3.459	3.393	3.404
	EXP2	0.38	3.333	3.460	3.413	3.401
	EXP3	0.56	3.309	3.465	3.435	3.401
	EXP4	0.75	3.290	3.461	3.446	3.397
	EXP5	1.09	3.250	3.442	3.452	3.379
	EXP6	2.19	3.083	3.351	3.422	3.278
	EXP7	2.78	3.052	3.342	3.416	3.262
R4	EXP1	0.20	3.315	3.418	3.396	3.376
	EXP2	0.30	3.289	3.424	3.417	3.376
	EXP3	0.45	3.269	3.426	3.436	3.375
	EXP4	0.60	3.266	3.435	3.450	3.382
	EXP5	0.88	3.210	3.410	3.446	3.352
	EXP6	1.75	3.070	3.321	3.398	3.257
	EXP7	2.23	3.043	3.306	3.382	3.237

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Data Availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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